

DRY-RUN GUIDE | percentage.java
Source: java\1st_belt\percentage.java

CORE BEHAVIOR

Reads obtained and total marks, casts before division to force floating-point arithmetic, multiplies by 100, and formats to two decimals.

REPRESENTATIVE INPUT / CALL

Input: '45 60'.

STEP TRACE

45.0/60=0.75; times 100=75.0; format with two fractional digits.

EXPECTED OUTPUT / RETURN

'75.00'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Challenge #5
003 |
004 | You are given the marks obtained by a student and the total (maximum) marks.
005 | Your task is to calculate the percentage scored and print it rounded to two decimal
    | places.
006 |
007 | The percentage is calculated using the formula:
008 | percentage = (marks_obtained / total_marks) * 100
009 |
010 | Use floating-point division, not integer division.
011 |
012 | Input Format:
013 | - A single line containing two space-separated integers marks_obtained and
    | total_marks
014 |
015 | Output Format:
016 | - Print the percentage scored, rounded to two decimal places.
017 | */
018 |
019 | import java.util.Scanner;
020 |
021 | public class Solution {
022 |     public static void main(String[] args) {
023 |         Scanner sc = new Scanner(System.in);
024 |         // Read marks_obtained and total_marks
025 |         int marks_obtained = sc.nextInt();
026 |         int total_marks = sc.nextInt();
027 |
028 |         // Calculate and print percentage to 2 decimal places
029 |         float percentage = ((float)marks_obtained / total_marks) * 100.0f;
030 |         System.out.printf("%.2f\n", percentage);
031 |     }
032 | }
033 |
```